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SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards *(BL2,BL6)*

Assessment Tool Weightage: 40%

38

Annexure A

Coding challenge

1. Develop a Python program that generates a Star Topology for a given number of hosts connected to a central switch. The program should accept the number of hosts as input and display the topology either in ASCII format or a graphical representation. For each host, print the host label, the name of the central switch, and the corresponding Cat6 UTP cable length connecting the host to the switch. Verify that all hosts are connected through the central switch and explain how the star topology facilitates reliable communication in a local area network.

2. Write a Python function named validate_segment(length_m) to verify whether a given **UTP cable segment** satisfies the IEEE 802.3 Ethernet standard, which specifies a maximum cable length of **100 meters**. The function should return **True** for valid cable lengths and return **False** along with an appropriate error message for invalid lengths exceeding the permissible limit. Demonstrate the correctness